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2 (1) $\lim_{x \to a} \dfrac{x^n - a^n}{x-a} = \lim_{x \to a} n \cdot x^{n-1}$

$= n \cdot a^{n-1}$

(2) $\lim_{x \to 1} \dfrac{(x-2)(\sqrt{x}-1)}{x^2+x-2} = \lim_{x \to 1} \dfrac{(x-2)(\sqrt{x}-1)(\sqrt{x}+1)}{(x+2)(x-1)(\sqrt{x}+1)}$

$= \lim_{x \to 1} \dfrac{(x-2)}{(x+2)(\sqrt{x}+1)}$

$= -\dfrac{1}{6}$ //

3 (3) $\lim_{x \to 0} \dfrac{1}{x}\left(\dfrac{1}{\sqrt{x+2}} - \dfrac{1}{\sqrt{2}}\right)$

$= \lim_{x \to 0} \dfrac{1}{x}\left(\dfrac{\sqrt{2}-\sqrt{x+2}}{\sqrt{2}\cdot\sqrt{x+2}}\right)$

$= \lim_{x \to 0} \dfrac{1}{x} \cdot \dfrac{(-x)}{\sqrt{2}\cdot\sqrt{x+2}\left(\sqrt{2}+\sqrt{x+2}\right)}$

$= \dfrac{-1}{\sqrt{2}\cdot\sqrt{2}\left(\sqrt{2}+\sqrt{2}\right)} = \dfrac{-1}{4\sqrt{2}}$

$= -\dfrac{\sqrt{2}}{8}$ //

(4) $\lim_{x \to 0} \dfrac{\sqrt{x+2}-\sqrt{2}}{x}$

$= \lim_{x \to 0} \dfrac{x}{x\left(\sqrt{x+2}+\sqrt{2}\right)}$

$= \lim_{x \to 0} \dfrac{1}{\sqrt{x+2}+\sqrt{2}}$

$= \dfrac{1}{2\sqrt{2}} = \dfrac{\sqrt{2}}{4}$ //

5 (3) $\lim_{t \to \infty} \left(\sqrt{t^2-1} - \sqrt{t^2+4}\right)$

$= \lim_{t \to \infty} \dfrac{-5}{\sqrt{t^2-1}+\sqrt{t^2+4}}$

$= 0$ //

(5) $\lim_{x \to 3-} \dfrac{3-x}{|x-3|} = \lim_{x \to 3-} \dfrac{3-x}{-(x-3)}$

$= \lim_{x \to 3-} \dfrac{3-x}{3-x} = 1$ //